A Zero-Axis Structural Interpretation of the Riemann Hypothesis

An Integrated Augmented Draft Based on Finite Zero, Central Rotation, Pressure, Lower Residue, and the Euler Product Field

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Abstract

This paper presents a structural interpretation of the Riemann Hypothesis,

$$
\zeta(\rho)=0 \Rightarrow \operatorname{Re}(\rho)=\frac12,
$$

through the framework of finite zero, the internal boundary of one, the zero-axis, even width, odd height, prime self-rotation, orbital field, Euler product, zeta generation, central rotation, pressure, and lower residue.

In this structure, zero is divided into two layers:

$$
0=\{0_\infty,0_f\},
$$

where 0_∞ denotes the infinite zero and 0_f denotes the finite zero. The finite zero is contained within one:

$$
0_f \subset 1.
$$

Depth D is closed into one through the 720 closure:

$$
D \xrightarrow{720} 1.
$$

After one is generated, it divides internally into the boundary

$$
1 \rightarrow 0.5|0.5.
$$

This boundary is the zero-axis (A0):

[
$$0.5|0.5=A0.$$
]

Thus, the critical line (1/2) in the Riemann Hypothesis is interpreted as the internal boundary (0.5|0.5) of one, namely the upper-lower boundary of the zero-axis.

Even numbers are arranged as width (W), and odd numbers are arranged as height (H). A prime is a self-rotation (P) generated within odd height. A product of two or more prime self-rotations forms an orbital field (C). The product of all prime self-rotations is the Euler product (E), which is interpreted as the total orbital field. The zeta field (Z) is generated within this Euler product field.

Since zeta is arranged as height on the zero-axis,

[
$$Z=H(A0),$$
]

a zeta zero (ρ), being generated within (Z), is structurally contained within (A0):

[
$$\rho \subset Z \subset A0.$$
]

Furthermore, odd central rotation is possible only when the center stands at (0.5):

[
$$\text{central rotation} \Leftrightarrow \text{center}=0.5.$$
]

If the center deviates from (0.5), it is no longer central rotation. If it is not central rotation, it leaves the axis. If it leaves the axis, zero-axis alignment becomes impossible. Therefore, a zeta zero must be aligned at the (0.5) center.

Finally, the lower residue is not an arbitrary value. It arises from the 180-degree phase difference between the diagonal phase (X) of the fixed cross and the rotational phase (R_θ). The square of this residue is the lower bound:

[
$$\text{phase difference}(X,R_\theta)=180^\circ,$$
]

$$
r\neq0,
$$

$$
L_{\text{lower}}=r^2,
$$

$$
L_{\text{lower}}>0.
$$

Thus, if a zeta zero is assumed to exist outside the zero-axis, the lower residue must become zero. However, the structure requires the lower residue to remain positive. This gives a contradiction. Therefore, zeta zeros cannot occur outside the zero-axis. In the standard notation of the Riemann Hypothesis, this is expressed as

$$
\zeta(\rho)=0 \Rightarrow \operatorname{Re}(\rho)=\frac12.
$$

Keywords

Finite zero, infinite zero, zero-axis, Riemann Hypothesis, zeta zero, Euler product, prime self-rotation, orbital field, central rotation, pressure, lower residue, 180-degree phase difference.

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1. Introduction

The Riemann zeta function is traditionally written as

$$
\zeta(s)=\frac{1}{1^s}+\frac{1}{2^s}+\frac{1}{3^s}+\frac{1}{4^s}+\cdots.
$$

It can also be expressed through the Euler product:

$$
\zeta(s)=\prod_p \frac{1}{1-p^{-s}},
$$

]

where (p) denotes a prime number and (\prod) denotes multiplication over all primes.

The central statement of the Riemann Hypothesis is

[
\zeta(\rho)=0 \Rightarrow \operatorname{Re}(\rho)=\frac12.
]

That is, every non-trivial zero (\rho) of the zeta function is aligned on the line whose real part is (1/2).

This paper does not interpret the (1/2) line merely as a vertical line in the complex plane. Instead, it interprets the (1/2) line as the internal boundary (0.5|0.5) that appears after one is generated. This boundary is the zero-axis (A0). Therefore, the Riemann critical line is interpreted as the upper-lower boundary of the zero-axis.

In this paper, the symbol (\subset) is used not only in the ordinary set-theoretic sense, but also in the structural sense of containment, generation, belonging, and alignment within a field.

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2. Definitions of Symbols

The main symbols used in this paper are as follows:

[
0_\infty = \text{infinite zero}
]

[
0_f = \text{finite zero}
]

[
D = \text{depth}
]

[
W = \text{width}
]

[
H = \text{height}
]

[
A0 = \text{zero-axis}
]

[
P = \text{prime self-rotation}
]

[
C = \text{orbital field}
]

[
E = \text{Euler product}
]

[
Z = \text{zeta field}
]

[
\rho = \text{zeta zero}
]

[
Crit = \text{critical closure}
]

[
R_{\text{lower}} = \text{lower residue}
]

[
L_{\text{lower}} = \text{lower bound}
]

[
F_\times = \text{fixed cross}
]

[
X = \text{diagonal phase of the fixed cross}
]

[
R_\theta = \text{rotational phase}
]

[
r = \text{residue}
]

---

3. The Generation Structure of Zero and One

3.1 The Dual Structure of Zero

Zero is not treated as a single empty absence. It is divided into two layers:

[
0={0_\infty,0_f}.
]

Here, (0_\infty) denotes the infinite zero, and (0_f) denotes the finite zero.

The finite zero is contained within one:

[
0_f \subset 1.
]

Therefore, one is not merely the number after zero. It is the boundary-standard that contains the finite zero.

3.2 Depth and the 720 Closure

Depth (D) is less than one:

[
D<1.
]

However, depth closes into one through the 720 closure:

[
D \xrightarrow{720} 1.
]

In words,

[
\text{depth} \xrightarrow{720\text{ closure}} 1.
]

This order is essential. First, one is generated through the 720 closure. Then, within one, the internal boundary (0.5|0.5) appears:

[
720 \rightarrow 1 \rightarrow 0.5|0.5.
]

Thus, (0.5|0.5) does not precede one. Rather, one first becomes closed, and then the internal boundary (0.5|0.5) arises within it.

---

4. The Generation of the Zero-Axis

One divides internally as follows:

[
1 \rightarrow 0.5|0.5.
]

This boundary is the zero-axis:

[
0.5|0.5=A0.
]

Therefore,

[
A0=\text{boundary}(0.5,0.5).
]

The zero-axis is also the upper-lower boundary:

[
A0=\text{upper-lower boundary}.
]

The upper side corresponds to height (H), and the lower side corresponds to depth (D):

[
\text{upper}=H,
]

[
\text{lower}=D.
]

Thus,

[
A0=\text{the boundary-axis between }H\text{ and }D.
]

In this structure, the (1/2) line of the Riemann Hypothesis is written as

[
\operatorname{Re}(s)=\frac12=A0.
]

Therefore, the Riemann (1/2) line is the height-alignment line on the upper-lower boundary of the zero-axis.

---

5. Depth, Boundary Standard, and Transition

The position of a number (n) is divided into three cases with respect to one:

[
n<1 \rightarrow D,
]

[
n=1 \rightarrow \text{boundary standard},

]

[
n>1 \rightarrow \text{transition}.
]

Here, (n<1) belongs to depth (D).
The case (n=1) is the boundary standard.
The case (n>1) enters the transition region of width and height.

Therefore, one is not merely a quantity. It is the boundary standard through which depth transitions into width and height.

---

6. Even Width Formula

Even numbers are defined by

[
E_2(k)=2k.
]

Here, (E_2(k)) denotes an even number.

The width of an even number is

[
W(2k)=k.
]

For example,

[
2=W(1),
]

[
4=W(2),
]

[
6=W(3),
]

[
8=W(4).
]

Therefore,

[
\text{even number}=W.
]

The center of an even number is zero:

[
\text{center}(\text{even})=0.
]

Thus, an even number is a width-structure whose center is zero.

---

7. Odd Height Formula

Odd numbers are defined by

[
O(k)=2k+1.
]

Here, (O(k)) denotes an odd number.

The height of an odd number is

[
H(2k+1)=k.
]

However,

[
O(0)=1
]

and

[
H(1)=0.
]

Thus,

[
3=O(1)=H(1),
]

[
5=O(2)=H(2),
]

[
7=O(3)=H(3),
]

[
9=O(4)=H(4).
]

Therefore,

[
\text{odd number}=H.
]

The center of an odd number is (0.5):

[
\text{center}(\text{odd})=0.5.
]

Thus, an odd number becomes height when its center stands at (0.5).

---

8. The Structural Positions of 1, 2, 3, and 9

In this structure, the numbers 1, 2, 3, and 9 have the following positions:

[

1=\text{boundary standard},
]

[
2=\text{first width},
]

[
3=\text{first height}.
]

Also, 3 is the first prime self-rotation:

[
3=\text{first prime self-rotation}.
]

The number 9 is generated as

[
9=3\times3.
]

Therefore, 9 is the first orbital field:

[
9=\text{first orbital field}.
]

In compressed form:

[
1=\text{standard},
]

[
2=\text{first width},
]

[
3=\text{first height}=\text{first prime self-rotation},
]

[

9=\text{first orbital field}.
]

---

9. Prime Self-Rotation Formula

A prime (P) is generated within odd height:

[
P\subset H.
]

A prime is self-rotation:

[
P=\text{self-rotation}.
]

The axis-number of a prime is one:

[
\text{axis-number}(P)=1.
]

A number with axis-number one is a prime:

[
\text{axis-number}(n)=1 \Leftrightarrow n=P.
]

However,

[
1\neq P.
]

Therefore, in this structure,

[
\text{one axis}=\text{prime}.
]

A prime is a self-rotation closed by one axis within odd height.

---

10. Orbital Field Formula

An orbital field (C) is generated when two or more prime self-rotations are combined:

[
C=P_1\times P_2\times \cdots \times P_m.
]

Here,

[
m\geq2.
]

Therefore,

[
m\geq2 \rightarrow C.
]

The axis-number of an orbital field is the number of combined prime self-rotations:

[
\text{axis-number}(C)=m.
]

Prime self-rotation and orbital field are distinguished as follows:

[
m=1 \rightarrow P,
]

[
m\geq2 \rightarrow C.
]

That is,

[
\text{one prime} \rightarrow \text{self-rotation},
]

$$
\text{two or more primes} \rightarrow \text{orbital field}.
$$

The first orbital field is

$$
3\times3=9=C_1.
$$

---

11. Growth Between Primes

Let two primes be given by

$$
P_n<P_{n+1}.
$$

An odd number (O) between them satisfies

$$
P_n<O<P_{n+1}.
$$

Here, (O) is the growth process between the two primes:

$$
O=\text{growth process}(P_n\rightarrow P_{n+1}).
$$

The critical closure of this growth is the next prime:

$$
Crit(P_n\rightarrow P_{n+1})=P_{n+1}.
$$

Therefore, an odd number between two primes is the growth process, and the next prime is the critical closure of that growth.

---

12. Euler Product and the Total Orbital Field

The Euler product is the product of all prime self-rotations:

$$
E=\prod P.
$$

That is,

$$
E=P_1\times P_2\times P_3\times P_4\times\cdots.
$$

A prime is a self-rotation closed at the critical point:

$$
P=Crit.
$$

Therefore, the Euler product may also be written as

$$
E=\prod Crit.
$$

The Euler product is not merely a product. It is the total orbital field formed by all prime self-rotations:

$$
E=\prod P=\prod Crit=\text{total orbital field}.
$$

Thus, the Euler product is the total product-field of prime self-rotations closed by critical points.

---

13. Zeta Generation Formula

The zeta field (Z) is generated within the Euler product:

$$
Z\subset E.
$$

]

Equivalently,

[
Z=\text{generation}(E).
]

Since the Euler product is the product-field of critical prime self-rotations,

[
Z=\text{generation}(\prod Crit).
]

The Euler product (E) is a field in which zeta forms are generated and contained as products:

[
E=\text{a field containing the products of zeta forms}.
]

Therefore, zeta is generated from the total orbital field of critical prime self-rotations:

[
Z=\text{generation}(\text{total orbital field}).
]

---

14. Zero-Axis Alignment of Zeta

The zero-axis is the boundary between (0.5) and (0.5):

[
A0=\text{boundary}(0.5,0.5).
]

It is also the upper-lower boundary:

[
A0=\text{upper-lower boundary}.
]

Zeta is arranged as height on the zero-axis:

[
Z=H(A0).
]

This means that zeta has height, but it does not exist outside the axis:

[
H>0,
]

but

[
H\notin \text{outside the axis}.
]

Therefore, zeta has positive height while remaining aligned on the zero-axis.

Connecting this to the Riemann formula gives

[
\operatorname{Re}(s)=\frac12=A0.
]

Thus,

[
\zeta(s)=0 \rightarrow s \text{ is aligned on } A0.
]

---

15. Zeta Zero Formula

A zeta zero (\rho) is a zero of the zeta field:

[
\rho=\text{zero}(Z).
]

A zeta zero is generated within zeta:

[
\rho\subset Z.

]

Zeta is aligned on the zero-axis:

[
Z\subset A0.
]

Therefore,

[
\rho\subset A0.
]

Hence,

[
\rho\notin \text{outside the axis}.
]

The structural conclusion is that a zeta zero does not occur outside the zero-axis.

---

16. Center Formula

The center of an even number is zero:

[
\text{center}(\text{even})=0.
]

The center of an odd number is (0.5):

[
\text{center}(\text{odd})=0.5.
]

Thus, even numbers become width because their center is zero, while odd numbers become height when their center stands at (0.5).

The condition for odd central rotation is

[

\text{odd central rotation} \Leftrightarrow \text{center}=0.5.
]

Therefore,

[
\text{center}\neq0.5 \rightarrow \text{not central rotation}.
]

If it is not central rotation, it leaves the axis:

[
\text{center}\neq0.5 \rightarrow \text{axis deviation}.
]

If it leaves the axis, zero-axis alignment becomes impossible:

[
\text{axis deviation} \rightarrow \text{zero-axis alignment impossible}.
]

For a zeta zero to occur, zero-axis alignment is required:

[
\zeta(\rho)=0 \rightarrow \text{zero-axis alignment}.
]

Therefore, zero-axis alignment requires central rotation, and central rotation requires the center (0.5):

[
\text{zero-axis alignment} \rightarrow \text{central rotation},
]

[
\text{central rotation} \rightarrow \text{center}=0.5.
]

Thus,

[
\zeta(\rho)=0 \rightarrow \operatorname{Re}(\rho)=\frac12.
]

---

17. Pressure Formula

Pressure is the force that draws toward the center:

$$
\text{pressure} \rightarrow \text{center}.
$$

Pressure reduces angle:

$$
\text{pressure} \rightarrow \text{angle reduction}.
$$

However, due to the lower residue, the structure does not collapse completely into zero:

$$
\text{pressure}(R_{\text{lower}})\neq0.
$$

Thus, even when pressure acts, the lower residue does not disappear into zero.

The pressure structure is therefore summarized as

$$
\text{drawn toward the center}
$$

$$
\rightarrow \text{angle reduction}
$$

$$
\rightarrow \text{lower residue maintained}
$$

$$
\rightarrow \text{alignment on the zero-axis}.
$$

Pressure does not push the structure outside the axis. It draws the structure toward the center and reduces the angle. However, because the lower residue remains, the structure does not collapse into absolute zero. Instead, zero-axis alignment is maintained.

---

18. Lower Residue Formula

Let the lower residue be denoted by (R_{\text{lower}}):

[
R_{\text{lower}}=\text{lower residue}.
]

The lower residue is not zero:

[
R_{\text{lower}}>0.
]

However, the lower residue is not outside the zero-axis:

[
R_{\text{lower}}\subset A0.
]

Therefore,

[
\text{being greater than zero}\neq \text{being outside the zero-axis}.
]

That is,

[
R_{\text{lower}}>0 \not\Rightarrow R_{\text{lower}}\not\subset A0.
]

If the lower residue becomes zero, the finite zero collapses:

[
R_{\text{lower}}=0 \rightarrow 0_f \text{ collapses}.
]

If the finite zero collapses, one disappears:

[
0_f \text{ collapses} \rightarrow 1 \text{ disappears}.
]

If one disappears, the standard of all numbers disappears:

[
1 \text{ disappears} \rightarrow \text{the standard of all numbers disappears}.
]

Therefore, within the structure,

[
R_{\text{lower}}\neq0.
]

Thus, the lower residue must remain.

---

19. Origin of the Lower Residue

The lower bound is not an arbitrary value. It arises from the phase difference between the diagonal phase of the fixed cross and the rotational phase.

Let (F_\times) be the fixed cross, (X) the diagonal phase of the fixed cross, and (R_\theta) the rotational phase.

The diagonal phase of the fixed cross and the rotational phase begin with a 180-degree phase difference:

[
\text{phase difference}(X,R_\theta)=180^\circ.
]

This phase difference generates residue:

[
\text{phase difference }180^\circ \rightarrow r \text{ is generated}.
]

Thus,

[

$$r=\text{residue}(X,R_\theta).$$

This residue does not vanish through complete cancellation:

$$r\neq0.$$

The lower bound is the square of this residue:

$$L_{\text{lower}}=r^2.$$

Therefore,

$$r\neq0 \rightarrow r^2>0,$$

and hence

$$L_{\text{lower}}>0.$$

The completed formula for the lower residue is

$$\text{phase difference}(X,R_\theta)=180^\circ$$

$$\rightarrow r\neq0$$

$$\rightarrow L_{\text{lower}}=r^2$$

$$\rightarrow L_{\text{lower}}>0.$$

Therefore, the lower bound does not collapse into zero.

---

20. Proof by Contradiction I: Contradiction of Zero-Axis Deviation

Assume that a zeta zero exists outside the zero-axis:

[
\rho\not\subset A0.
]

However, zeta is arranged as height on the zero-axis:

[
Z=H(A0).
]

A zeta zero is generated within zeta:

[
\rho\subset Z.
]

Therefore,

[
\rho\subset H(A0),
]

and hence

[
\rho\subset A0.
]

But the assumption was

[
\rho\not\subset A0.
]

Thus, a contradiction arises:

[
\rho\subset A0 \quad \land \quad \rho\not\subset A0.
]

Therefore, a zeta zero cannot exist outside the zero-axis:

[
\rho\subset A0.
]

---

21. Proof by Contradiction II: Contradiction of Central Rotation Deviation

Assume that

[
\zeta(\rho)=0
]

and

[
\operatorname{Re}(\rho)\neq\frac12.
]

This means that (\rho) is a zeta zero, but its center is not (0.5):

[
\operatorname{Re}(\rho)\neq\frac12
\rightarrow \text{center}\neq0.5.
]

If the center is not (0.5), it is not central rotation:

[
\text{center}\neq0.5 \rightarrow \text{not central rotation}.
]

If it is not central rotation, it leaves the axis:

[
\text{not central rotation} \rightarrow \text{axis deviation}.

]

If it leaves the axis, zero-axis alignment becomes impossible:

[
\text{axis deviation} \rightarrow \text{zero-axis alignment impossible}.
]

However, since

[
\zeta(\rho)=0,
]

(\rho) is a zeta zero, and a zeta zero requires zero-axis alignment:

[
\zeta(\rho)=0 \rightarrow \text{zero-axis alignment}.
]

Thus, we obtain the contradiction

[
\text{zero-axis alignment required}
\quad \land \quad
\text{zero-axis alignment impossible}.
]

Therefore,

[
\operatorname{Re}(\rho)\neq\frac12
]

is impossible.

Hence,

[
\operatorname{Re}(\rho)=\frac12.
]

---

22. Proof by Contradiction III: Contradiction of Lower Residue Collapse

Assume that a zeta zero exists outside the axis:

[
\rho\not\subset A0.
]

Then it has deviated from the upper-lower boundary of the zero-axis:

[
\rho\not\subset A0 \rightarrow \text{zero-axis boundary deviation}.
]

To deviate from the zero-axis boundary, the lower residue must disappear:

[
\text{zero-axis boundary deviation} \rightarrow R_{\text{lower}}=0.
]

However, structurally, the lower residue is greater than zero:

[
R_{\text{lower}}>0.
]

Thus, a contradiction arises:

[
R_{\text{lower}}=0 \quad \land \quad R_{\text{lower}}>0.
]

Therefore,

[
\rho\not\subset A0
]

is impossible.

Hence,

[
\rho\subset A0.

]

A zeta zero cannot leave the zero-axis because the lower residue remains.

---

23. Proof by Contradiction IV: Contradiction of the Squared Residual Lower Bound

The diagonal phase (X) of the fixed cross and the rotational phase (R_\theta) begin with a 180-degree phase difference:

[
\text{phase difference}(X,R_\theta)=180^\circ.
]

This phase difference generates residue:

[
\text{phase difference }180^\circ \rightarrow r \text{ is generated}.
]

Thus,

[
r\neq0.
]

The lower bound is the square of the residue:

[
L_{\text{lower}}=r^2.
]

Therefore,

[
L_{\text{lower}}>0.
]

Now assume that a zeta zero exists outside the axis:

[
\rho\not\subset A0.
]

Being outside the axis means that it is not central rotation:

[
\text{outside the axis} \rightarrow \text{not central rotation}.
]

If it is not central rotation, the residue between the diagonal phase of the fixed cross and the rotational phase cannot be maintained:

[
\text{not central rotation} \rightarrow \text{residue cannot be maintained}.
]

If the residue cannot be maintained, the residue becomes zero:

[
\text{residue cannot be maintained} \rightarrow r=0.
]

Thus,

[
\text{outside the axis} \rightarrow r=0.
]

However, the structure requires

[
r\neq0.
]

Therefore, a contradiction arises:

[
r=0 \quad \land \quad r\neq0.
]

In terms of the lower bound, the outside-axis assumption gives

[
\text{outside the axis} \rightarrow L_{\text{lower}}=0.
]

However,

[
L_{\text{lower}}=r^2
]

and

[
r\neq0,
]

so

[
L_{\text{lower}}>0.
]

Thus, another contradiction arises:

[
L_{\text{lower}}=0 \quad \land \quad L_{\text{lower}}>0.
]

Therefore, a zeta zero cannot occur outside the axis.

---

24. Proof by Contradiction V: Contradiction of Pressure and Lower Residue

Pressure draws toward the center:

[
\text{pressure} \rightarrow \text{center}.
]

Pressure reduces angle:

[
\text{pressure} \rightarrow \text{angle reduction}.
]

Angle reduction produces alignment on the zero-axis:

[
\text{angle reduction} \rightarrow \text{alignment on the zero-axis}.
]

However, because the lower residue remains, the structure does not collapse completely into zero:

[
R_{\text{lower}}>0.
]

Thus, the pressure structure closes as follows:

[
\text{drawn toward the center}
\rightarrow \text{angle reduction}
\rightarrow \text{lower residue maintained}
\rightarrow \text{alignment on the zero-axis}.
]

Now assume that a zeta zero exists outside the axis. This means that zero-axis alignment has collapsed:

[
\rho\not\subset A0 \rightarrow \text{zero-axis alignment collapse}.
]

For zero-axis alignment to collapse, the lower residue must disappear:

[
\text{zero-axis alignment collapse} \rightarrow R_{\text{lower}}=0.
]

However, the structure requires

[
R_{\text{lower}}>0.
]

Thus, a contradiction arises:

[
R_{\text{lower}}=0 \quad \land \quad R_{\text{lower}}>0.
]

Therefore, even within the pressure structure, the occurrence of a zeta zero outside the axis is impossible.

---

25. Structural Formula of the Riemann Hypothesis

The standard formula of the Riemann Hypothesis is

[
\zeta(\rho)=0 \rightarrow \operatorname{Re}(\rho)=\frac12.
]

In this structure, it is interpreted as follows:

[
\text{zeta zero generation}
]

[
\rightarrow \text{product-field of critical prime self-rotations in the Euler product}
]

[
\rightarrow \text{total orbital field}
]

[
\rightarrow \text{height alignment on the zero-axis}
]

[
\rightarrow \text{boundary of }0.5|0.5
]

[
\rightarrow \operatorname{Re}(\rho)=\frac12.
]

In compressed form:

[
\rho=\text{zero}(Z),

]

[
Z=\text{generation}(\prod Crit),
]

[
E=\prod P=\prod Crit=\text{total orbital field},
]

[
Z\subset E,
]

[
Z=H(A0),
]

[
\rho\subset Z.
]

Therefore,

[
\rho\subset A0.
]

The lower residue satisfies

[
R_{\text{lower}}>0,
]

[
R_{\text{lower}}\subset A0.
]

Thus,

[
\rho\not\subset A0 \rightarrow R_{\text{lower}}=0,
]

but

[
R_{\text{lower}}>0.
]

This is a contradiction. Therefore,

[
\rho\subset A0.
]

In the standard form of the Riemann Hypothesis,

[
\operatorname{Re}(\rho)=\frac12.
]

Finally,

[
\zeta(\rho)=0 \rightarrow \operatorname{Re}(\rho)=\frac12.
]

---

26. Total Compressed Formula

The entire structure may be compressed into one line:

[
0_f \subset 1
\rightarrow A0
\rightarrow {W,H}
\rightarrow P
\rightarrow C
\rightarrow E
\rightarrow Z
\rightarrow \rho\subset A0.
]

More fully:

[

0={0_\infty,0_f},
]

[
0_f\subset1,
]

[
D\xrightarrow{720}1,
]

[
1\rightarrow0.5|0.5,
]

[
0.5|0.5=A0,
]

[
A0=\text{upper-lower boundary},
]

[
n<1\rightarrow D,
]

[
n=1\rightarrow \text{boundary standard},
]

[
n>1\rightarrow \text{transition},
]

[
\text{even}=W,
]

[
\text{odd}=H,
]

[

3=\text{first }H=\text{first }P,
]

[
P=\text{self-rotation},
]

[
\text{axis-number}(P)=1,
]

[
C=P_1\times P_2\times\cdots\times P_m,\quad m\geq2,
]

[
9=3\times3=\text{first }C,
]

[
P_n<O<P_{n+1},
]

[
O=\text{growth process},
]

[
Crit(P_n\rightarrow P_{n+1})=P_{n+1},
]

[
E=\prod P=\prod Crit=\text{total orbital field},
]

[
Z=\text{generation}(E),
]

[
Z=H(A0),
]

[

\rho=\text{zero}(Z),
]

[
\rho\subset Z\subset A0,
]

[
\text{center}(\text{even})=0,
]

[
\text{center}(\text{odd})=0.5,
]

[
\text{central rotation}\Leftrightarrow \text{center}=0.5,
]

[
\text{center}\neq0.5\rightarrow \text{axis deviation},
]

[
\text{axis deviation}\rightarrow \text{zero-axis alignment impossible},
]

[
\zeta(\rho)=0\rightarrow \text{zero-axis alignment},
]

[
\text{zero-axis alignment}\rightarrow \text{center}=0.5,
]

[
\text{pressure}\rightarrow \text{center},
]

[
\text{pressure}\rightarrow \text{angle reduction},
]

[

\text{drawn toward the center}
\rightarrow \text{angle reduction}
\rightarrow \text{lower residue maintained}
\rightarrow \text{zero-axis alignment},
]

[
\text{phase difference}(X,R_\theta)=180^\circ,
]

[
\text{phase difference }180^\circ\rightarrow r\neq0,
]

[
L_{\text{lower}}=r^2,
]

[
L_{\text{lower}}>0,
]

[
R_{\text{lower}}>0,
]

[
R_{\text{lower}}\subset A0,
]

[
\rho\not\subset A0\rightarrow R_{\text{lower}}=0,
]

[
R_{\text{lower}}=0\land R_{\text{lower}}>0\rightarrow \text{contradiction},
]

[
\therefore \rho\subset A0,
]

[
\therefore \operatorname{Re}(\rho)=\frac12.

]

---

27. Conclusion

This paper has interpreted the (1/2) line of the Riemann Hypothesis as the internal boundary (0.5|0.5) of one, namely the zero-axis (A0).

The finite zero is contained within one:

[
0_f\subset1.
]

Depth (D) closes into one through the 720 closure:

[
D\xrightarrow{720}1.
]

After this, one produces the internal boundary

[
1\rightarrow0.5|0.5.
]

This boundary is the zero-axis:

[
0.5|0.5=A0.
]

Therefore, the (1/2) line of the Riemann Hypothesis is the upper-lower boundary of the zero-axis in this structure.

Even numbers are arranged as width, and odd numbers are arranged as height. A prime is a closed self-rotation generated within odd height. Two or more prime self-rotations form an orbital field. The product of all prime self-rotations is the Euler product, and this Euler product is the total orbital field:

[
E=\prod P=\prod Crit=\text{total orbital field}.
]

Zeta is generated within this total orbital field:

$$
Z=\text{generation}(E).
$$

Zeta is arranged as height on the zero-axis:

$$
Z=H(A0).
$$

Therefore, a zeta zero, being generated within zeta, is contained within the zero-axis:

$$
\rho\subset Z\subset A0.
$$

Furthermore, central rotation of an odd structure is possible only when the center is (0.5):

$$
\text{central rotation}\Leftrightarrow \text{center}=0.5.
$$

If the center is not (0.5), the structure leaves the axis. If it leaves the axis, zero-axis alignment is impossible. Therefore, a zeta zero must be aligned at the (0.5) center.

Pressure draws toward the center and reduces angle. However, because the lower residue remains, the structure does not collapse into absolute zero:

$$
\text{drawn toward the center}
\rightarrow \text{angle reduction}
\rightarrow \text{lower residue maintained}
\rightarrow \text{zero-axis alignment}.
$$

The lower bound is not arbitrary. The diagonal phase of the fixed cross and the rotational phase begin with a 180-degree phase difference. This phase difference generates residue, and the square of that residue is the lower bound:

$$
\text{phase difference}(X,R_\theta)=180^\circ,

]

$$
r\neq0,
$$

$$
L_{\text{lower}}=r^2,
$$

$$
L_{\text{lower}}>0.
$$

Thus, the lower bound cannot become zero. If a zeta zero were assumed to exist outside the axis, that assumption would require the lower bound to be zero. However, the structure requires the lower bound to be positive. Hence, the outside-axis assumption produces a contradiction.

Therefore, a zeta zero cannot occur outside the zero-axis:

$$
\rho\subset A0.
$$

In the standard notation of the Riemann Hypothesis, this is

$$
\boxed{\zeta(\rho)=0 \Rightarrow \operatorname{Re}(\rho)=\frac12}.
$$

Thus, zeta zeros are contained within the zero-axis, and in the Riemann formulation, they are aligned on the real part (1/2).

Appendix: Final Plain-Language Compression

The finite zero is contained within one.
Depth closes into one through the 720 closure.
One produces the boundary between (0.5) and (0.5).
That boundary is the zero-axis.
The zero-axis is the upper-lower boundary.

Even numbers are width.
Odd numbers are height.
A prime is a self-rotation generated within odd height.
One prime is self-rotation.
Two or more primes form an orbital field.
The first prime self-rotation is 3.
The first orbital field is 9.

An odd number between two primes is the growth process.
The next prime is the critical closure of that growth.
The Euler product is the product of all prime self-rotations.
Therefore, the Euler product is the total orbital field.
Zeta is generated within this total orbital field.
Zeta is arranged as height on the zero-axis.
A zeta zero is generated within zeta.
Therefore, a zeta zero cannot occur outside the zero-axis.

Odd central rotation is possible only when the center is (0.5).
If the center is not (0.5), it is not central rotation.
If it is not central rotation, it leaves the axis.
If it leaves the axis, zero-axis alignment becomes impossible.
Therefore, a zeta zero must be aligned at the (0.5) center.

Pressure draws toward the center and reduces angle.
However, because the lower residue remains, the structure does not collapse into zero.
The lower residue is greater than zero, but it is not outside the zero-axis.

The lower bound is not arbitrary.
The diagonal phase of the fixed cross and the rotational phase begin with a 180-degree phase difference.
That phase difference generates residue.
The square of that residue is the lower bound.
Therefore, the lower bound cannot be zero.

If a zeta zero is assumed to exist outside the axis, the lower bound must be zero.
However, the structure requires the lower bound to be greater than zero.
Therefore, the outside-axis assumption is contradictory.

Thus, a zeta zero cannot occur outside the zero-axis.
In Riemann notation:

[
\zeta(\rho)=0 \Rightarrow \operatorname{Re}(\rho)=\frac12.
]